PRINCIPLES OF PROGRAMMING USING C

Course Code: PPC18 **Credits: 2:0:1**

Pre-requisites: - **Contact Hours: 28L+14P**

Course Coordinator: Dr. Ganeshayya Shidaganti

Course Content

Unit I

Introduction to computers: Computer Systems, Computing Environment, Computer Languages, Creating and Running Programs, System Development. **Introduction to C programming:** Background, Structure of C Program, Identifiers, Data Types, Variables, Constants, Input/ Output.Fornattiing Input/Output. Library Functions, Single Character Input –The getchar() , Single Character Output-The putchar(),

- Pedagogy/Course delivery tools: Chalk and talk, Power point presentation, Videos
- Links: https://nptel.ac.in/courses/106/105/106105171/ MOOC courses can be adopted for more clarity in understanding the topics and verities of problem solving methods

Unit II

Operators and Expressions: Arithmetic operators, Unary operators, Relational and Logical Operators, Assignment Operators, Conditional Operators. Expressions, Precedence and Associativity, Evaluating Expressions, Type Conversion. **Selection - Making Decision:** Two Way Selection, Multiway Selection, **Repetition:** Concepts of a loop, Pretest and Post-test Loops, Initialization and Updating, Loops in C: The while loop, the do. while loop, The for loop, The break and continue statement, the goto statement.

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Unit III

Arrays: Concepts, Using arrays in C: Declaration and Definition, Accessing Elements in Array, Storing Values in Arrays, Processing an array, Sorting and Searching. Two Dimensional Arrays, Multidimensional Arrays. **Functions in C:** A brief Overview, Defining a Function, User Defined Functions, Function Prototypes, Passing Arguments to a Function, Scope-global and local; Recursion.

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Unit IV

Storage Classes: automatic variables, external (global) variables, static variables **Strings:** String Concepts, Defining a String, Declaring Strings, Initializing String, Arrays of Strings, String Manipulations Functions **Pointers:** Pointer Concepts, Pointer Declaration and Definition, Initialization of Pointer Variables, Passing Pointers to a Function, Pointers and Array, Pointers and Charter Array.

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Unit V

Derived Types: Enumerated Types: Declaring an Enumerated Type, Operations on Enumerated Types, Initializing Enumerated Constants. **Structures and Unions:** Structure Type Declaration, Initialization, Accessing Structures, Structures Containing Arrays, Array of Structures. Unions and Structures. **File Handling:** Using Binary Files, Reading and Writing a Data File.

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Course Outcomes (COs):

At the end of the course the student will be able to:

1. Identify the basic elements of Computing Systems and C Programming Constructs.
2. Demonstrate the use of Operators & Expressions, Decision Making and Looping Statements.
3. Explore Arrays and User-Defined Functions in Implementing Solutions to Real world Problems
4. Illustrate the usage of Storage Classes, Strings and Pointers in Problem Solving.
5. Demonstrate the use of Modular Programming Constructs involving Files, Structure & Unions.

Lab Component:

Sl. No	QUESTIONS
1.	**Creating and Running Simple C Programs:** • C-Program to calculate the sum of three numbers / C-Program to demonstrate a Simple Calculator. • C-Program to calculate the area and circumference of a circle using PI as a defined constant. • C-Program to convert temperature given in Celsius to Fahrenheit and Fahrenheit to Celsius • C-Program to compute the roots of a quadratic equation by accepting the coefficients.

2.	**Creating and Running C Programs on Expressions :** • C-Program to calculate quotient and reminder of two numbers. (Page: 125) • C-Program to evaluate two complex expressions. (Page:113) • C-Program to demonstrate automatic and type casting of numeric types(Page: 117 and 119) • C-Program to calculate the total sales given the unit price, quantity, discount and tax rate(Page: 130) • C-Program to calculate a student's average score for a course with 4 quizzes, 2 midterms and a final. The quizzes are weighted 30%, the midterms 40% and the final 30%. (Page: 131)
3.	**Creating and Running C Programs on Making Decision:** • C-Program to determine the use of the character classification functions found in c-type library. (Page:267) • C-Program to read a test score, calculate the grade for the score and print the grade. (Page: 259) • C-Program to uses a menu to allow the user to add, multiply, subtract and divide two numbers using switch case. (Page: 277) • C-Program to read the name of the user, number of units consumed and print out the charges. An electricity board charges the following rates for the use of electricity: o For the first 200 units 80 paise per unit o For the next 100 units 90 paise per unit o Beyond 300 units Rs 1 per unit. All users are charged a minimum of Rs. 100 as meter charge. If the total amount is more than Rs 400, then an additional surcharge of 15% of total amount is charged.
4.	**Creating and Running C Programs on Repetition or Loops:** • C-Program to print a number series from 1 to a user-specified limit in the form of a right triangle (Page: 328) • C-Program to print the number and sum of digits in an integer. (Page:332) • C-Program to calculate the factorial of a number using for loop/ Recursion(Page:351) • C-Program to calculate nth Fibonacci number. (Page:355) • C-Program to convert binary to a decimal number (Page: 335)
5.	**Creating and Running C Programs on One Dimensional Arrays:** • C-Program to print square of index and print it. (Page:471) • C-Program to calculate average of the number in an array. (Page:477) • C-Program to sort the list using bubble sort. (Page:495) • C-Program to search an ordered list using binary search. (Page:508)
6.	**Creating and Running C Programs on Two Dimensional Arrays:** • C-Program to perform addition of two matrices. • C-Program to perform multiplication of two matrices. • C-Program to find transpose of the given matrices. • C-Program to find row sum and column sum and sum of all elements in a matrix. • C-Program initialize/fill all the diagonal elements of a matrix with zero and print.

7.	**Creating and Running C Programs on User Defined Functions:** • C-program to read a number, Find its factorial using function with argument and with return type. • C-Program to read two number, Find its GCD and LCM using function with arguments and without return type. • C-Program to read a number, Find whether it is a palindrome or not using function without argument and with return type. • C-Program to read a number, Find whether it is prime number or not using function without arguments and without return type.
8.	**Creating and Running C Programs on Strings:** • C-program read two strings, Combine them without using string built-in functions. • C-program read two strings, Compare them without using string built-in functions. • C-program read two strings, concatenate them without using string built-in functions. • C Program to Check if the Substring is Present in the Given String. • C-program to demonstrate built-in sting functions like strlen(), strcpy(), strcmp(), strcat().
9.	**Creating and Running C Programs on Storage Classes and Pointers:** • C-program to show the use of auto and static variable. (Page:1108) • C-program to add two numbers using pointers. / C-program to swap two numbers using pointers. • C-program to show how the same pointer can point to different data variable. (Page:571)/ C-program to show the use of different pointers point to the same variable (Page:572) • C-Program to read an array of elements, Compute its sum using pointers.
10.	**Creating and Running C Programs on Derived Types and and Unions:** • C-Program to print selected TV stations for our cable TV systems. (Page: 751) • C-Program to demonstrate union of short int and two char (Page:783)
11.	**Creating and Running C Programs on Structures:** • C Program to read employee details (name, salary, address) and print the same using structure. • C-Program to read marks of three students in 3 subjects. Calculate the total marks scored, student wise and subject wise using structure.
12.	**Creating and Running C Programs on Files:** • C-Program to demonstrate function fread()/fscanf() • C-Program to demonstrate function fwrite()/fprintf()

Text Book:

1. **Behrouz A. Forouzan, Richard F. Gilberg -** Computer science a structured programming approach using C, Cengage Learning, and ISBN: 9788131503638, 8131503631, 3rd edition, 2007.

Reference Books:

1. **E. Balaguruswamy -** Programming in ANSI C, Tata McGraw-Hill, 7th edition,
2. **Reema Thareja** - Computer fundamentals and programming in C, Oxford University, 2nd edition, 2017.
3. **Brian W. Kernighan and Dennis M. Ritchie** - The 'C' Programming Language, Prentice Hall of India.

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	**Marks**	**Course outcomes attained**
Internal Test-I	30	CO1, CO2, CO3
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal test shall be taken for 30 marks		
Other components		
Lab Component Evaluation	20	CO1, CO2, CO3, CO4, CO5
Semester-End Examination (SEE)	100	CO1, CO2, CO3, CO4, CO5